In [1]:
```
#get plots for signficant trends in MHW properties along new cal coast
#get timeseries of %eez spatial extent
#get total number of days in hot season and cold season around new caledoni
a
```

In [3]:
```
from cartopy import config
import cartopy.crs as ccrs
import xarray as xr
import geopandas as gpd
import matplotlib as mpl
import matplotlib.pyplot as plt
import matplotlib.colors as mcolors
import numpy as np
import pandas as pd
import os
```

In [4]:
```
data_crs = ccrs.PlateCarree(central_longitude=0)

# But you want the map to include the dateline.
proj_crs = ccrs.PlateCarree(central_longitude=180)
```

In [5]:
```
# eez and country boundaries
eez = gpd.read_file("/media/shilpa/Expansion/for_sophie_18_april_2024/World
_EEZ_v11_20191118/eez_v11.shp")
```

In [6]:
```
ncd = eez[eez["TERRITORY1"] == "New Caledonia"]
```

In [6]:
```
# there are three types of files:
# 1. all events at a particular location
# 2. mean, median, std, trend, significance at a particular location
# 3. timeseries for the spatial extent plots
```

In [7]:
```
# We will use the second type of file; mean, median, std, trend, significan
ce at a particular location to make the following plot:
# significant trends in number of events, duration, mean and max intensitie
s , total annual days and total annual cumaltive intensities
```

In [8]:
```
directory = '/media/shilpa/Expansion/for_sophie_18_april_2024/coastal_newca
l/'

# Get a list of CSV files starting with 'samoa'
coastal_files = [file for file in os.listdir(directory) if file.startswith
("New Caledonia_mean_trends")]

# Initialize an empty list to store dataframes
dfs = []

# Loop through each CSV file, read it into a dataframe, and append it to th
e list
for file in coastal_files:
    filepath = os.path.join(directory, file)
    df = pd.read_csv(filepath)
    dfs.append(df)

# Concatenate all dataframes into one
result = pd.concat(dfs, ignore_index=True)

# Print or do whatever you want with the concatenated dataframe
#print(len(result)) # there are 709 coastal point for New Caledonia
```

```
709
```

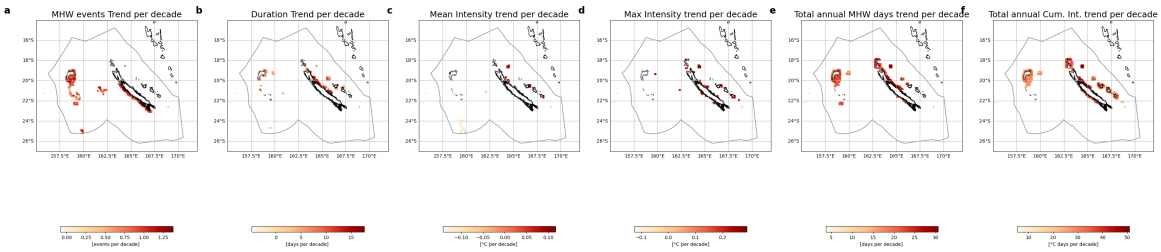
In [9]:
```
gdf = gpd.GeoDataFrame(result, geometry=gpd.points_from_xy(result.longitud
e, result.latitude))
```

In [11]:
```
# filter to keep only significant trends....its just like masking in xarray
, it will apply the filter to the whole dataframe

filtered_gdf_count = gdf[gdf['sig_trend_per_decade_count'] == 1].copy()

filtered_gdf_duration = gdf[gdf['sig_trend_per_decade_duration'] == 1].copy
()

filtered_gdf_mean_intensity = gdf[gdf['sig_trend_per_decade_intensity_mea
n'] == 1].copy()

filtered_gdf_max_intensity = gdf[gdf['sig_trend_per_decade_intensity_max_ma
x'] == 1].copy()

filtered_gdf_annual_days = gdf[gdf['sig_trend_per_decade_total_days'] ==
1].copy()

filtered_gdf_annual_icum = gdf[gdf['sig_trend_per_decade_total_icum'] ==
1].copy()
```

In [56]:
```
def easy_plot_unevencolorbars_(data,column,i,s,name,legend_kwds, vmin,vma
x):

    norm = mpl.colors.Normalize(vmin=vmin, vmax=vmax)
    cbar = plt.cm.ScalarMappable(norm=norm, cmap='OrRd')


    data.plot(column,ax=axs[i],marker = 's',s = s,transform=data_crs,alpha
= 1,cmap='OrRd', norm = norm,
            legend=True,legend_kwds=legend_kwds)

    axs[i].coastlines()

    ncd.boundary.plot(ax=axs[i], color="black",alpha = 0.4,transform=data_c
rs)

    gl = axs[i].gridlines(draw_labels=True)
    gl.top_labels = False
    gl.right_labels = False
    axs[i].set_xlim(-25,-8)
    axs[i].set_ylim(-27,-14)
    axs[i].set_title(name, fontsize=18)
```

In [57]:
```
# only ploting significant trend

nrows = 1
ncols = 6

fig, axs = plt.subplots(nrows=nrows,ncols=ncols,subplot_kw={'projection': p
roj_crs},figsize=(34,15))



easy_plot_unevencolorbars_(data=filtered_gdf_count,column= 'count_trend_per
_decade',name='MHW events Trend per decade',i=0,s=10,
                           vmin = gdf.count_trend_per_decade.min(),vmax = gd
f.count_trend_per_decade.max(),legend_kwds={'label': "[events per decade]
",
                                            'orientation': "horizontal",'shrin
k': 0.7})

easy_plot_unevencolorbars_(data=filtered_gdf_duration,column= 'duration_tre
nd_per_decade',name='Duration Trend per decade',i=1,s=10,
                           vmin = gdf.duration_trend_per_decade.min(),vmax =
gdf.duration_trend_per_decade.max(),legend_kwds={'label': "[days per decad
e] ",
                                            'orientation': "horizontal",'shrin
k': 0.7})

easy_plot_unevencolorbars_(data=filtered_gdf_mean_intensity,column= 'intens
ity_mean_trend_per_decade',name='Mean Intensity trend per decade',i=2, s=1
0,
                           vmin = gdf.intensity_mean_trend_per_decade.min
(),vmax = gdf.intensity_mean_trend_per_decade.max(),legend_kwds={'label':
"[°C per decade] ",
                                            'orientation': "horizontal",'shrin
k': 0.7})

easy_plot_unevencolorbars_(data=filtered_gdf_max_intensity,column= 'intensi
ty_max_max_trend_per_decade',name='Max Intensity trend per decade',i=3, s=1
0,
                           vmin = gdf.intensity_max_max_trend_per_decade.min
(),vmax = gdf.intensity_max_max_trend_per_decade.max(),legend_kwds={'labe
l': "[°C per decade]",
                                            'orientation': "horizontal",'shrin
k': 0.7})

easy_plot_unevencolorbars_(data=filtered_gdf_annual_days,column= 'total_day
s_trend_per_decade',name='Total annual MHW days trend per decade',i=4, s=1
0,
                           vmin = gdf.total_days_trend_per_decade.min(),vmax
= gdf.total_days_trend_per_decade.max(),legend_kwds={'label': "[days per de
cade] ",
                                            'orientation': "horizontal",'shrin
k': 0.7})

easy_plot_unevencolorbars_(data=filtered_gdf_annual_icum,column= 'total_icu
m_trend_per_decade',name='Total annual Cum. Int. trend per decade',i=5, s=1
```

```
0,
                          vmin = gdf.total_icum_trend_per_decade.min(),vmax
= gdf.total_icum_trend_per_decade.max(),legend_kwds={'label': "[°C days per
decade]",
                                                 'orientation': "horizontal",'shrin
k': 0.7})

axs[0].text(-0.2, 1.05, 'a', transform=axs[0].transAxes, fontsize=20, fontw
eight='bold')
axs[1].text(-0.2, 1.05, 'b', transform=axs[1].transAxes, fontsize=20, fontw
eight='bold')
axs[2].text(-0.2, 1.05, 'c', transform=axs[2].transAxes, fontsize=20, fontw
eight='bold')
axs[3].text(-0.2, 1.05, 'd', transform=axs[3].transAxes, fontsize=20, fontw
eight='bold')
axs[4].text(-0.2, 1.05, 'e', transform=axs[4].transAxes, fontsize=20, fontw
eight='bold')
axs[5].text(-0.2, 1.05, 'f', transform=axs[5].transAxes, fontsize=20, fontw
eight='bold')

plt.tight_layout()
#plt.savefig('/home/shilpa/glory_mat_analysis/coastal_points/newcal_coast_t
rends_only_signf2.png',bbox_inches='tight')
```



In [ ]:
```
# We will use the third type of file; timeseries for the spatial extent plo
ts to make the following plot:
# timeseries of  % new caledonia coastal and EEZ in MHW state
```

In [14]:
```
# glorys linear original
glorys_linear = pd.read_csv('/media/shilpa/Expansion/for_sophie_18_april_20
24/combined_linear_glorys.csv')
sorted_gl = glorys_linear.sort_values('time_index')
sorted_gl['time'] = pd.to_datetime(sorted_gl['time'])

# noaa full
noaa_1981_2023 = pd.read_csv('/media/shilpa/Expansion/for_sophie_18_april_2
024/noaa_%eez_spatial_extent.csv')
sorted_n_1981_2023 = noaa_1981_2023.sort_values('time_index')
sorted_n_1981_2023['time'] = pd.to_datetime(sorted_n_1981_2023['time'])

# noaa full macroscale
noaa_macro_scale = pd.read_csv('/media/shilpa/Expansion/for_sophie_18_april
_2024/%eez_spatialextent_25_more.csv')
noaa_macro_scale_1981_2023 = noaa_macro_scale.sort_values('time_index')
noaa_macro_scale_1981_2023['time'] = pd.to_datetime(noaa_macro_scale_1981_2
023['time'])


# glorys full macroscale
glorys_macro_scale = pd.read_csv('/media/shilpa/Expansion/for_sophie_18_apr
il_2024/glorys_spatial_macroscale.csv')
glorys_macro_scale_1993_2019 = glorys_macro_scale.sort_values('time_index')
glorys_macro_scale_1993_2019['time'] = pd.to_datetime(glorys_macro_scale_19
93_2019['time'])
```

In [16]:
```
# replace zeros with nans
df_replacedgl = sorted_gl.replace(0, np.nan)
df_replacedn_1981_2023 = sorted_n_1981_2023.replace(0, np.nan)
noaa_macro_scale_1981_2023_replaced = noaa_macro_scale_1981_2023.replace
(0,np.nan)
glorys_macro_scale_1993_2019_replaced = glorys_macro_scale_1993_2019.replac
e(0,np.nan)
```

In [26]:
```
# lets get the coastal information
directory = '/media/shilpa/Expansion/for_sophie_18_april_2024/coastal_newca
l'
coastal_files = [file for file in os.listdir(directory) if file.startswith
("New Caledonia_spatial")]

# Read all CSV files and concatenate into a single DataFrame
dfs = []
for file in coastal_files:
    # Handle file names with spaces by adding double quotes around the file
path
    file_path = os.path.join(directory, file)
    df = pd.read_csv(file_path)
    dfs.append(df)

# Concatenate all DataFrames
concatenated_df = pd.concat(dfs, ignore_index=True)

# Assuming the column containing presence and absence data is named 'presen
ce'
# Group by 'time' (assuming it's the column name) and calculate the percent
age of presence for each day
coastal_percentage = concatenated_df.groupby('time')['spatial_extent'].mean
() * 100
```

In [20]:
```
presence_df = pd.DataFrame({'time': coastal_percentage.index, 'coastal_perc
entage': coastal_percentage.values})
```

In [21]:
```
start_date = '1993-01-01'
end_date = '2023-10-24'

# Create datetime range
date_range = pd.date_range(start=start_date, end=end_date)
print(len(date_range))
```

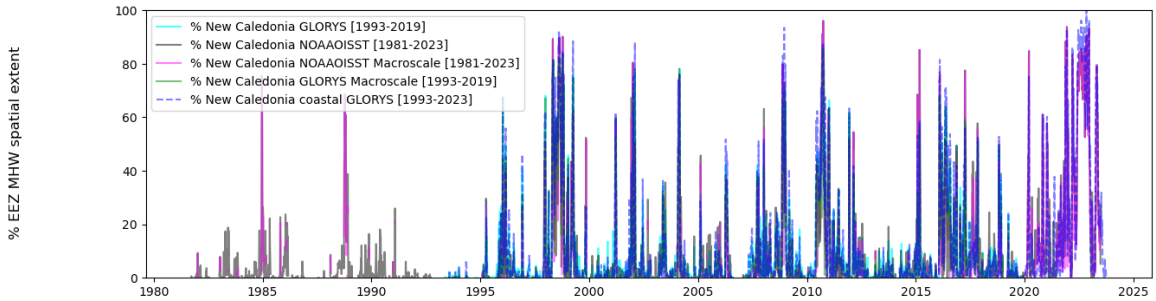
```
11254
```

In [22]:
```
presence_df = pd.DataFrame({'time': coastal_percentage.index,'time_dt':date
_range.values, 'coastal_percentage': coastal_percentage.values})
```

In [23]:
```
coastal = presence_df.replace(0,np.nan)
```

In [24]:
```
fig, ax = plt.subplots(figsize=(15, 4))# constrained_layout=True)

ax.plot(df_replacedgl.time,df_replacedgl['New_Caledonia_%spatial_extent'],
color='cyan',alpha = 0.7,linestyle='solid',label='% New Caledonia GLORYS [1
993-2019] ')
ax.plot(df_replacedn_1981_2023.time,df_replacedn_1981_2023['New_Caledonia
_%spatial_extent'], color='black',alpha = 0.5,linestyle='solid',label='% Ne
w Caledonia NOAAOISST [1981-2023]')
ax.plot(noaa_macro_scale_1981_2023_replaced.time,noaa_macro_scale_1981_2023
_replaced['New_Caledonia_%spatial_extent'], color='magenta',alpha = 0.5,lin
estyle='solid',label='% New Caledonia NOAAOISST Macroscale [1981-2023]')
ax.plot(glorys_macro_scale_1993_2019_replaced.time,glorys_macro_scale_1993_
2019_replaced['New_Caledonia_%spatial_extent'], color='green',alpha = 0.5,l
inestyle='solid',label='% New Caledonia GLORYS Macroscale [1993-2019]')
ax.plot(coastal.time_dt,coastal['coastal_percentage'], color='blue',alpha =
0.5,linestyle='dashed',label='% New Caledonia coastal GLORYS [1993-2023]')

ax.legend()
ax.set_ylim(0, 100)
fig.supylabel('% EEZ MHW spatial extent')
#fig.savefig('/home/shilpa/glory_mat_analysis/coastal_points/timeseries_%ee
z_spatial_extent_newcaledonia.png')
```

Out[24]:
```
Text(0.02, 0.5, '% EEZ MHW spatial extent')
```

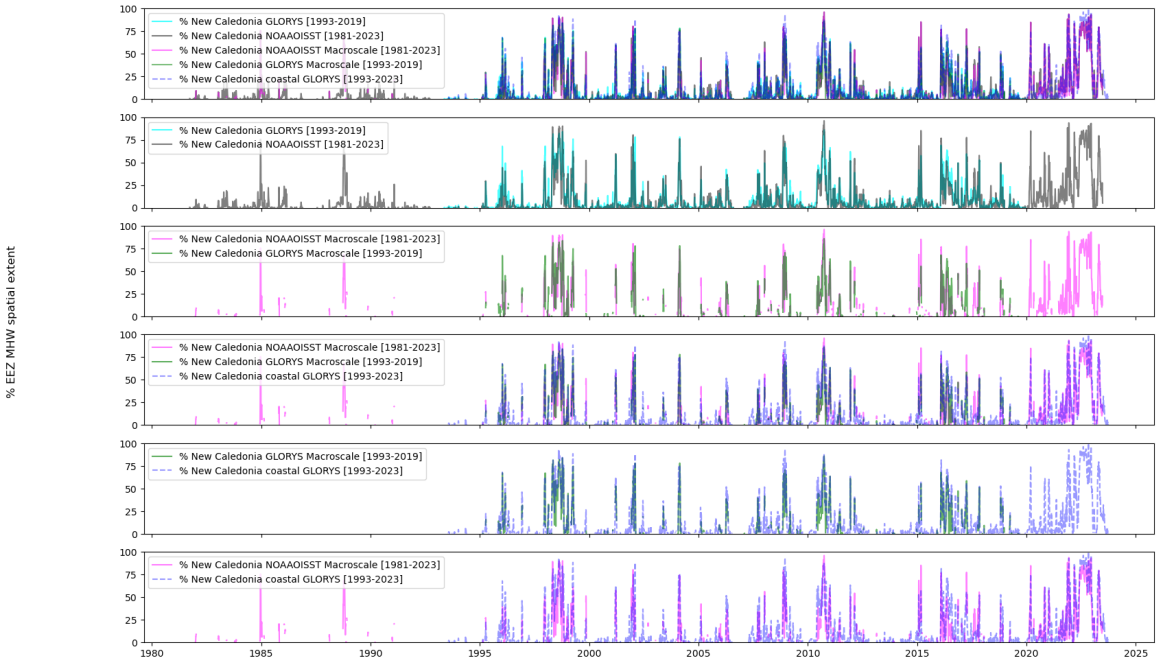


In [25]:
```
fig, axs = plt.subplots(6, 1,sharex=True, figsize=(19, 12))# constrained_la
yout=True)

ax = axs[0]
ax.plot(df_replacedgl.time,df_replacedgl['New_Caledonia_%spatial_extent'],
color='cyan',alpha = 0.7,linestyle='solid',label='% New Caledonia GLORYS [1
993-2019] ')
ax.plot(df_replacedn_1981_2023.time,df_replacedn_1981_2023['New_Caledonia
_%spatial_extent'], color='black',alpha = 0.5,linestyle='solid',label='% Ne
w Caledonia NOAAOISST [1981-2023]')
ax.plot(noaa_macro_scale_1981_2023_replaced.time,noaa_macro_scale_1981_2023
_replaced['New_Caledonia_%spatial_extent'], color='magenta',alpha = 0.5,lin
estyle='solid',label='% New Caledonia NOAAOISST Macroscale [1981-2023]')
ax.plot(glorys_macro_scale_1993_2019_replaced.time,glorys_macro_scale_1993_
2019_replaced['New_Caledonia_%spatial_extent'], color='green',alpha = 0.5,l
inestyle='solid',label='% New Caledonia GLORYS Macroscale [1993-2019]')
ax.plot(coastal.time_dt,coastal['coastal_percentage'], color='blue',alpha =
0.4,linestyle='dashed',label='% New Caledonia coastal GLORYS [1993-2023]')
ax.legend()
ax.set_ylim(0, 100)

ax = axs[1]
ax.plot(df_replacedgl.time,df_replacedgl['New_Caledonia_%spatial_extent'],
color='cyan',alpha = 0.7,linestyle='solid',label='% New Caledonia GLORYS [1
993-2019] ')
ax.plot(df_replacedn_1981_2023.time,df_replacedn_1981_2023['New_Caledonia
_%spatial_extent'], color='black',alpha = 0.5,linestyle='solid',label='% Ne
w Caledonia NOAAOISST [1981-2023]')
#ax.plot(noaa_macro_scale_1981_2023_replaced.time,noaa_macro_scale_1981_202
3_replaced['New_Caledonia_%spatial_extent'], color='magenta',alpha = 0.5,li
nestyle='solid',label='% New Caledonia NOAAOISST Macroscale [1981-2023]')
#ax.plot(glorys_macro_scale_1993_2019_replaced.time,glorys_macro_scale_1993
_2019_replaced['New_Caledonia_%spatial_extent'], color='green',alpha =
0.5,linestyle='solid',label='% New Caledonia GLORYS Macroscale [1993-201
9]')
#ax.plot(coastal.time_dt,coastal['coastal_percentage'], color='blue',alpha
= 0.5,linestyle='dashed',label='% New Caledonia coastal GLORYS [1993-202
3]')
ax.legend()
ax.set_ylim(0, 100)

ax = axs[2]
#ax.plot(df_replacedgl.time,df_replacedgl['New_Caledonia_%spatial_extent'],
color='cyan',alpha = 0.7,linestyle='solid',label='% New Caledonia GLORYS [1
993-2019] ')
#ax.plot(df_replacedn_1981_2023.time,df_replacedn_1981_2023['New_Caledonia
_%spatial_extent'], color='black',alpha = 0.5,linestyle='solid',label='% Ne
w Caledonia NOAAOISST [1981-2023]')
ax.plot(noaa_macro_scale_1981_2023_replaced.time,noaa_macro_scale_1981_2023
_replaced['New_Caledonia_%spatial_extent'], color='magenta',alpha = 0.5,lin
estyle='solid',label='% New Caledonia NOAAOISST Macroscale [1981-2023]')
ax.plot(glorys_macro_scale_1993_2019_replaced.time,glorys_macro_scale_1993_
2019_replaced['New_Caledonia_%spatial_extent'], color='green',alpha = 0.6,l
inestyle='solid',label='% New Caledonia GLORYS Macroscale [1993-2019]')
#ax.plot(coastal.time_dt,coastal['coastal_percentage'], color='blue',alpha
```

```
= 0.5,linestyle='dashed',label='% New Caledonia coastal GLORYS [1993-202
3]')
ax.legend()
ax.set_ylim(0, 100)

ax = axs[3]
#ax.plot(df_replacedgl.time,df_replacedgl['New_Caledonia_%spatial_extent'],
color='cyan',alpha = 0.7,linestyle='solid',label='% New Caledonia GLORYS [1
993-2019] ')
#ax.plot(df_replacedn_1981_2023.time,df_replacedn_1981_2023['New_Caledonia
_%spatial_extent'], color='black',alpha = 0.5,linestyle='solid',label='% Ne
w Caledonia NOAAOISST [1981-2023]')
ax.plot(noaa_macro_scale_1981_2023_replaced.time,noaa_macro_scale_1981_2023
_replaced['New_Caledonia_%spatial_extent'], color='magenta',alpha = 0.5,lin
estyle='solid',label='% New Caledonia NOAAOISST Macroscale [1981-2023]')
ax.plot(glorys_macro_scale_1993_2019_replaced.time,glorys_macro_scale_1993_
2019_replaced['New_Caledonia_%spatial_extent'], color='green',alpha = 0.6,l
inestyle='solid',label='% New Caledonia GLORYS Macroscale [1993-2019]')
ax.plot(coastal.time_dt,coastal['coastal_percentage'], color='blue',alpha =
0.4,linestyle='dashed',label='% New Caledonia coastal GLORYS [1993-2023]')
ax.legend()
ax.set_ylim(0, 100)


ax = axs[4]
#ax.plot(df_replacedgl.time,df_replacedgl['New_Caledonia_%spatial_extent'],
color='cyan',alpha = 0.7,linestyle='solid',label='% New Caledonia GLORYS [1
993-2019] ')
#ax.plot(df_replacedn_1981_2023.time,df_replacedn_1981_2023['New_Caledonia
_%spatial_extent'], color='black',alpha = 0.5,linestyle='solid',label='% Ne
w Caledonia NOAAOISST [1981-2023]')
#ax.plot(noaa_macro_scale_1981_2023_replaced.time,noaa_macro_scale_1981_202
3_replaced['New_Caledonia_%spatial_extent'], color='magenta',alpha = 0.5,li
nestyle='solid',label='% New Caledonia NOAAOISST Macroscale [1981-2023]')
ax.plot(glorys_macro_scale_1993_2019_replaced.time,glorys_macro_scale_1993_
2019_replaced['New_Caledonia_%spatial_extent'], color='green',alpha = 0.6,l
inestyle='solid',label='% New Caledonia GLORYS Macroscale [1993-2019]')
ax.plot(coastal.time_dt,coastal['coastal_percentage'], color='blue',alpha =
0.4,linestyle='dashed',label='% New Caledonia coastal GLORYS [1993-2023]')
ax.legend()
ax.set_ylim(0, 100)

ax = axs[5]
#ax.plot(df_replacedgl.time,df_replacedgl['New_Caledonia_%spatial_extent'],
color='cyan',alpha = 0.7,linestyle='solid',label='% New Caledonia GLORYS [1
993-2019] ')
#ax.plot(df_replacedn_1981_2023.time,df_replacedn_1981_2023['New_Caledonia
_%spatial_extent'], color='black',alpha = 0.5,linestyle='solid',label='% Ne
w Caledonia NOAAOISST [1981-2023]')
ax.plot(noaa_macro_scale_1981_2023_replaced.time,noaa_macro_scale_1981_2023
_replaced['New_Caledonia_%spatial_extent'], color='magenta',alpha = 0.5,lin
estyle='solid',label='% New Caledonia NOAAOISST Macroscale [1981-2023]')
#ax.plot(glorys_macro_scale_1993_2019_replaced.time,glorys_macro_scale_1993
_2019_replaced['New_Caledonia_%spatial_extent'], color='green',alpha =
0.5,linestyle='solid',label='% New Caledonia GLORYS Macroscale [1993-201
9]')
```

```
ax.plot(coastal.time_dt,coastal['coastal_percentage'], color='blue',alpha =
0.4,linestyle='dashed',label='% New Caledonia coastal GLORYS [1993-2023]')
ax.legend()
ax.set_ylim(0, 100)

fig.supylabel('% EEZ MHW spatial extent')
#fig.savefig('/home/shilpa/glory_mat_analysis/coastal_points/timeseries_%ee
z_spatial_extent_newcaledonia.png')
```

Out[25]: Text(0.02, 0.5, '% EEZ MHW spatial extent')



In [15]:
```
# We will use the third type of file; timeseries for the spatial extent plo
ts to make the following plot:
# total number of days in hot season and cold season around new caledonia
```

In [27]:
```
start_date = '1993-01-01'
end_date = '2023-10-24'

# Create datetime range
date_range = pd.date_range(start=start_date, end=end_date)
print(len(date_range))
```

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In [44]:
```
# lets get the coastal information
directory = '/media/shilpa/Expansion/for_sophie_18_april_2024/coastal_newca
l'
coastal_files = [file for file in os.listdir(directory) if file.startswith
("New Caledonia_spatial")]

# Read all CSV files and concatenate into a single DataFrame

# hot season

hot_daily_events = []
hot_lat = []
hot_lon = []

for file in coastal_files:
    # Handle file names with spaces by adding double quotes around the file
path
    file_path = os.path.join(directory, file)
    df = pd.read_csv(file_path)

    # use the first value of lat and lon because all values are the same...
this file is for only one location
    lat = df['latitude'].iloc[0]
    lon = df['longitude'].iloc[0]
    hot_lat.append(lat)
    hot_lon.append(lon)

    df['time_dt'] = date_range.values
    df['time_dt'] = pd.to_datetime(df['time_dt'])

    # Extract the month component
    df['Month'] = df['time_dt'].dt.month

    # Filter the DataFrame for months November (11) to April (4)
    filtered_df = df[(df['Month'] >= 11) | (df['Month'] <= 4)]

    # Optionally, you can drop the 'Month' column if you don't need it anym
ore
    filtered_df = filtered_df.drop(columns=['Month'])

    filtered_df_count = filtered_df[filtered_df['spatial_extent'] == 1].cop
y() # only keep lines with value of 1 for spatail extent

    #len(filtered_df_count) # gives the number of days in hot season with M
HW
    hot_daily_events.append(len(filtered_df_count))

hot_df = pd.DataFrame({'count_': hot_daily_events, 'hot_lat':hot_lat, 'hot_
lon': hot_lon})

hot_gdf = gpd.GeoDataFrame(hot_df, geometry=gpd.points_from_xy(hot_df.hot_l
on, hot_df.hot_lat))


# cold season
```

```
cold_daily_events = []
cold_lat = []
cold_lon = []

for file in coastal_files:
    # Handle file names with spaces by adding double quotes around the file
path
    file_path = os.path.join(directory, file)
    df = pd.read_csv(file_path)

    # use the first value of lat and lon because all values are the same...
this file is for only one location
    lat = df['latitude'].iloc[0]
    lon = df['longitude'].iloc[0]
    cold_lat.append(lat)
    cold_lon.append(lon)

    df['time_dt'] = date_range.values
    df['time_dt'] = pd.to_datetime(df['time_dt'])

    # Extract the month component
    df['Month'] = df['time_dt'].dt.month

    # Filter the DataFrame for months November (11) to April (4)
    # Filter the DataFrame for months May (5) to October (10)
    filtered_df = df[(df['Month'] >= 5) & (df['Month'] <= 10)]

    # Optionally, you can drop the 'Month' column if you don't need it anym
ore
    filtered_df = filtered_df.drop(columns=['Month'])

    filtered_df_count = filtered_df[filtered_df['spatial_extent'] == 1].cop
y() # only keep lines with value of 1 for spatail extent

    #len(filtered_df_count) # gives the number of days in hot season with M
HW
    cold_daily_events.append(len(filtered_df_count))

cold_df = pd.DataFrame({'count_': cold_daily_events, 'cold_lat':cold_lat,
'cold_lon': cold_lon})

cold_gdf = gpd.GeoDataFrame(cold_df, geometry=gpd.points_from_xy(cold_df.co
ld_lon, cold_df.cold_lat))
```

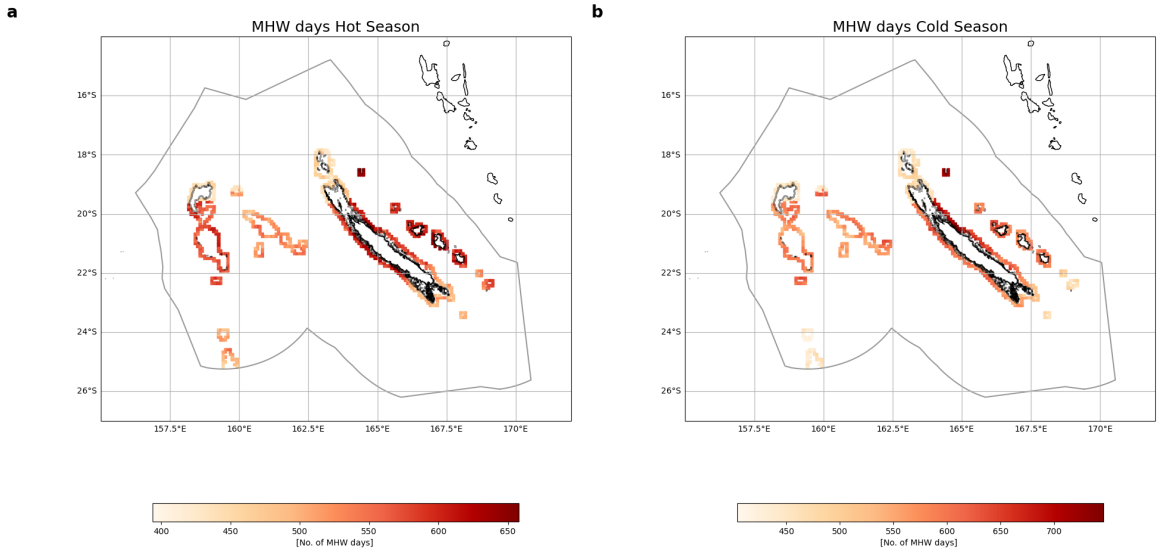
In [45]:
```
hot_gdf
```

Out[45]:

	count_	hot_lat	hot_lon	geometry
0	595	-19.583334	158.583328	POINT (158.58333 -19.58333)
1	452	-25.166666	159.916672	POINT (159.91667 -25.16667)
2	471	-25.083334	159.916672	POINT (159.91667 -25.08333)
3	489	-25.083334	159.833328	POINT (159.83333 -25.08333)
4	447	-25.166666	159.416672	POINT (159.41667 -25.16667)
...	...	...	...	...
704	437	-17.833334	162.833328	POINT (162.83333 -17.83333)
705	447	-17.833334	163.000000	POINT (163.00000 -17.83333)
706	431	-17.833334	163.083328	POINT (163.08333 -17.83333)
707	461	-17.833334	162.916672	POINT (162.91667 -17.83333)
708	438	-17.916666	162.750000	POINT (162.75000 -17.91667)

709 rows × 4 columns

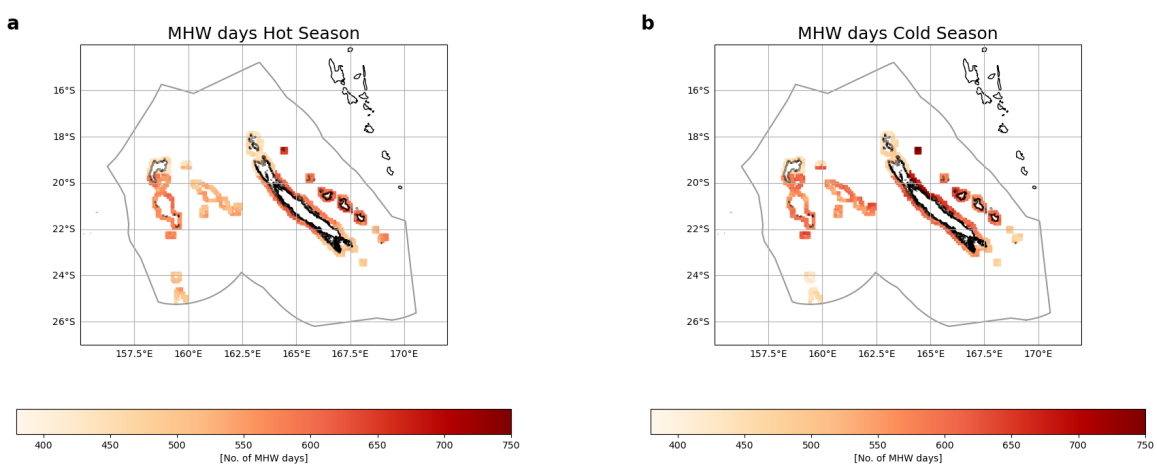
In [79]:
```
nrows = 1
ncols = 2

fig, axs = plt.subplots(nrows=nrows,ncols=ncols,subplot_kw={'projection': p
roj_crs},figsize=(20,10))



easy_plot_unevencolorbars_(data=hot_gdf,column= 'count_',name='MHW days Hot
Season',i=0, s = 10,
                           vmin = hot_gdf.count_.min(),vmax = hot_gdf.count
_.max(),legend_kwds={'label': "[No. of MHW days] ",
                                        'orientation': "horizontal",'shrin
k': 0.7})

easy_plot_unevencolorbars_(data=cold_gdf,column= 'count_',name='MHW days Co
ld Season',i=1, s = 10,
                           vmin = cold_gdf.count_.min(),vmax = cold_gdf.coun
t_.max(),legend_kwds={'label': "[No. of MHW days] ",
                                        'orientation': "horizontal",'shrin
k': 0.7})


axs[0].text(-0.2, 1.05, 'a', transform=axs[0].transAxes, fontsize=20, fontw
eight='bold')
axs[1].text(-0.2, 1.05, 'b', transform=axs[1].transAxes, fontsize=20, fontw
eight='bold')


plt.tight_layout()
#plt.savefig('/home/shilpa/glory_mat_analysis/coastal_points/newcal_coast_t
rends_only_signf2.png',bbox_inches='tight')
```



In [80]:
```
vmin = hot_gdf.count_.min()
vmax = hot_gdf.count_.max()
print( "hot_vmin =", vmin)
print( "hot_vmax =", vmax)

vmin = cold_gdf.count_.min()
vmax = cold_gdf.count_.max()
print( "cold_vmin =", vmin)
print( "cold_vmax =", vmax)
```

```
hot_vmin = 394
hot_vmax = 658
cold_vmin = 404
cold_vmax = 746
```

In [84]:
```
# change min and max colors to 380 and 750 ..so color bar is consistent.
# change marker size maybe

nrows = 1
ncols = 2

fig, axs = plt.subplots(nrows=nrows, ncols=ncols, subplot_kw={'projection':
proj_crs}, figsize=(20, 7))

# Plot for hot season
plot_hot = easy_plot_unevencolorbars_(data=hot_gdf, column='count_', name
='MHW days Hot Season', i=0, s=10,
                                      vmin=380, vmax=750,
                                      legend_kwds={'label': "[No. of MHW da
ys]",
                                                   'orientation': "horizont
al",
                                                   'shrink': 0.7})

# Plot for cold season
plot_cold = easy_plot_unevencolorbars_(data=cold_gdf, column='count_', name
='MHW days Cold Season', i=1, s=10,
                                       vmin=380, vmax=750,
                                       legend_kwds={'label': "[No. of MHW d
ays]",
                                                    'orientation': "horizon
tal",
                                                    'shrink': 0.7})

# Add text markers
axs[0].text(-0.2, 1.05, 'a', transform=axs[0].transAxes, fontsize=20, fontw
eight='bold')
axs[1].text(-0.2, 1.05, 'b', transform=axs[1].transAxes, fontsize=20, fontw
eight='bold')

plt.tight_layout()
#plt.show()
#plt.savefig('/home/shilpa/glory_mat_analysis/coastal_points/newcal_coast_h
ot_cold_days.png',bbox_inches='tight')
```




```
<Figure size 640x480 with 0 Axes>
```

In [ ]: